Q.1 Create a class named Student with attributes:

> name
>
> age
>
> course
>
> Create an object and print all the details.

In [1]:
```
class Student:
    Student_name="Shakshi"
    Student_age=18
    Student_course="B.Tech"

College=Student()

print(College.Student_name)
print(College.Student_age)
print(College.Student_course)
```

```
Shakshi
18
B.Tech
```

Q.2 Create a class named Car with attributes:

> brand
>
> model
>
> price
>
> Create two objects and display their details.

In [ ]:
```
class Car:
    pass

Car1=Car()
Car1.brand="Ferrari"
Car1.model="SF90 Stradale"
Car1.price="₹750,00,000"

Car2=Car()
Car2.brand="Toyota"
Car2.model = "Corolla"
Car2.price = "2000000"

print(Car1.brand)
print(Car1.model)
print(Car2.price)
print(Car2.brand)
print(Car2.model)
print(Car2.price)
```

```
Ferrari
SF90 Stradale
2000000
Toyota
Corolla
2000000
```

Q.3 Create a class named Employee and use a constructor (**init**) to initialize:

> employee_id
>
> name
>
> salary
>
> Print the employee details.

In [ ]:
```
class Employee:

    def __init__(self,employee_id ,name,salary):
        self.employee_id=employee_id
        self.name =name
        self.salary =salary

        print(f"{self.employee_id}")
        print(f"{self.name}")
        print(f"{self.salary}")


st=Employee(employee_id=123,name="sakshi",salary="200000")
```

```
123
sakshi
200000
```

Q.4 Create a class named Rectangle with attributes:

> length width Create a method to calculate and print the area of the rectangle.

In [3]:
```
class Rectangle:

    def __init__(self,length,width):
        self.length=length
        self.width=width

    def area(self):
        print("Area of Rectangle =", self.length * self.width)

r1=Rectangle(10,4)
r1.area()
```

```
Area of Rectangle = 40
```

Q.5 Create a class named Circle with an attribute

> radius Create a method that calculates and prints the area of the circle.

In [2]:
```
class Circle:

    def __init__(self,radius):
        self.radius=radius

    def area(self):
        return 3.14 * self.radius * self.radius

c1 = Circle(5)

print("Radius:-",c1.radius)
print("Area of the circle:-",c1.area())
```

```
Radius:- 5
Area of the circle:- 78.5
```

Q.6 Create a class named BankAccount with:

> account_holder
>
> balance
>
> Create methods: deposit(amount)
>
> withdraw(amount)
>
> Display the updated balance after each transaction.

In [3]:
```
class BankAccount:

    def __init__(self,account_holder,balance):
          self.account_holder=account_holder
          self.balance=balance

    def deposit(self,amount):
         self.balance=self.balance + amount
         print("Balance after deposit:",self.balance)

    def withdraw(self,amount):
         self.balance=self.balance - amount
         print("Balance after withdraw:",self.balance)

acc1=BankAccount("Sakshi",5000)

acc1.deposit(600)
acc1.withdraw(100)
```

```
Balance after deposit: 5600
Balance after withdraw: 5500
```

Q.7 Create a class named Animal with a method sound(). Create a child class Dog that overrides the sound() method and prints "Bark".

In [4]:
```
class Animal:

    def sound(self):
        print("animal make a sound")

class Dog(Animal):

    def sound(self):
        print("Bark")

d=Dog()
d.sound()
```

```
Bark
```

Q.8 Create a class named Person with attributes:

> name
>
> age
>
> Create a child class Student with an additional attribute: roll_number
>
> Display all the details using an object of the Student class.

In [10]:
```
class Person:
    def __init__(self,name,age):
        self.name=name
        self.age=age


class Student(Person):
        def __init__(self, name, age,roll_number):
         super().__init__(name, age)
         self.roll_number=roll_number

st=Student("Sakshi","18","42")

print("Name:-",st.name)
print("Age:-",st.age)
print("Roll number:-",st.roll_number)
```

```
Name:- Sakshi
Age:- 18
Roll number:- 42
```

Q.9 Create a class named Calculator with methods:

> add()
>
> subtract()

> multiply()
>
> divide()
>
> Take two numbers as input and perform all operations.

In [3]:
```
class Calculator:
    def add(self,a,b):
        print("Addition:-",a+b)

    def subtract(self,a,b):
        print("Subtraction:-",a-b)

    def multiply(self,a,b):
        print("Multiplication:-",a*b)

    def divide(self,a,b):
        if b!=0:
            print("Division:-",a/b)
        else:
            print("Division is not possible(cannot divide by zero)")

num1=int(input("Enter first number:-"))
num2=int(input("enter your second number:-"))

calc=Calculator()

calc.add(num1,num2)
calc.subtract(num1,num2)
calc.multiply(num1,num2)
calc.divide(num1,num2)
```

```
Addition:- 9
Subtraction:- -1
Multiplication:- 20
Division:- 0.8
```

Q.10. Create a class named LibraryBook with attributes:

> book_name
>
> author
>
> price
>
> Create a method display_book_info() that prints all book details.
>
> Create at least three book objects and display their information.

In [4]:
```
class LibraryBook:
    def __init__(self,book_name,author,price):

       self.book_name=book_name
       self.author=author
       self.price=price
```

```python
    def display_book_info(self):
        print("Book_name:-",self.book_name)
        print("Author:-",self.author)
        print("Price:-",self.price)
        print()

book1 = LibraryBook("Python Programming", "John Smith", 450)
book2 = LibraryBook("Data Science", "Alice Brown", 600)
book3 = LibraryBook("Machine Learning", "David Lee", 750)

book1.display_book_info()
book2.display_book_info()
book3.display_book_info()
```

```
Book_name:- Python Programming
Author:- John Smith
Price:- 450

Book_name:- Data Science
Author:- Alice Brown
Price:- 600

Book_name:- Machine Learning
Author:- David Lee
Price:- 750
```